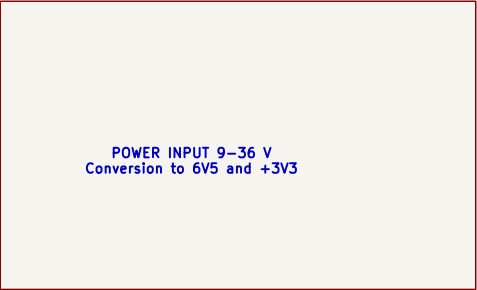


AUTOMOTIVE BLE–CAN MODULE

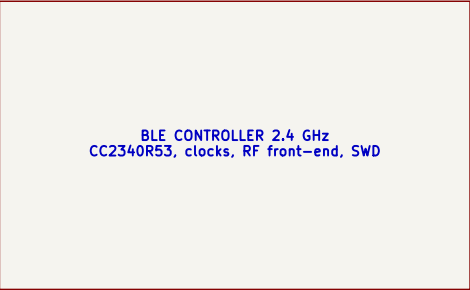
Vehicle CAN bus to Bluetooth LE bridge | 9–36 V input | 1S Li-ion backup
Automotive BLE–CAN Module | Rev A | September 2026

01_POWER_INPUT



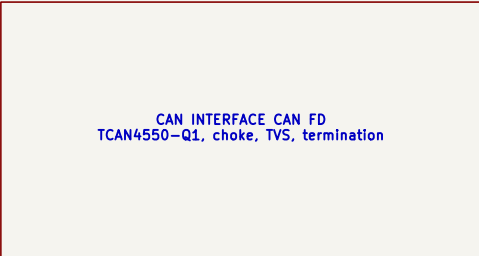
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02_MCU_BLE_Controller



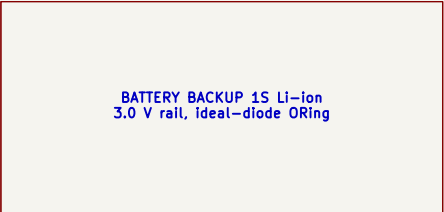
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03_CAN_Interface



File: 03_CAN_Interface.kicad_sch

04_Power_Back–up



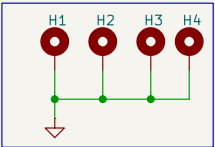
File: 04_Power_Back–up.kicad_sch

SHEET INDEX	
01	POWER INPUT 9–36 V protection, 6.49 V buck, 3.3 V LDO
02	BLE CONTROLLER CC2340R53, clocks, RF front–end, SWD
03	CAN INTERFACE TCAN4550–Q1, CMC, TVS, split termination
04	BATTERY BACKUP 1S pack input, 3.0 V LDO, ideal–diode ORing

KEY DEVICES	
U201	CC2340R53E0RKPR BLE 5.x MCU, 512 KB flash, QFN–40
U301	TCAN4550RGYRQ1 CAN FD controller + transceiver, SPI
U101	LMR38010S0DDARQ1 80 V buck, PFM, spread spectrum
U102	TPS7B8133QDGNRQ1 40 V LDO, 2.7 uA Iq
U402	TPS7E8130QDBVRQ1 40 V LDO, 2.8 uA Iq
U401/3/4	LM66100QDCKRQ1 ideal diode / reverse–current block
L301	ACT45B–101–2P CAN common–mode choke

POWER TREE
J101 9–36 V -> F101 -> D101 TVS / D102 reverse -> R101 damping
-> U101 LMR38010–Q1 buck +6V5_CAN (TCAN4550 VSUP)
-> U102 TPS7B8133–Q1 LDO ... +3V3_VEH
J401 1S pack -> F401 -> U401 RCB -> U402 TPS7E8130–Q1 ... +3V0_BAT
U403 / U404 LM66100–Q1 ideal–diode OR +3V3_SYS (MCU + VIO)
Vehicle rail has priority by the 3.3 V / 3.0 V offset; battery takes over automatically on vehicle loss. CAN is unavailable in backup.

Mounting Holes (M2)



Fiducial Markers



BOARD / PROCESS
4–layer FR4, 1.58 mm: F.Cu signal / In1_GND / In2_power / B.Cu GND
49.5 x 56.9 mm, single–sided assembly, HAL lead–free
Conformal coating REQUIRED – input spacing designed to IPC–2221 B4
RF feed is 50 ohm controlled impedance – see Output_Files/FAB_NOTES.md

Design period: 2026–09–24 to 2026–09–30 | Rev A
4–layer FR4, 1.58 mm, 49.5 x 56.9 mm | single–sided assembly
CC2340R53 BLE | TCAN4550–Q1 CAN FD | 9–36 V input | 1S Li–ion backup
Automotive BLE–CAN Module
SHIVAM TIWARI

Sheet: /
File: BLE–CAN–module.kicad_sch

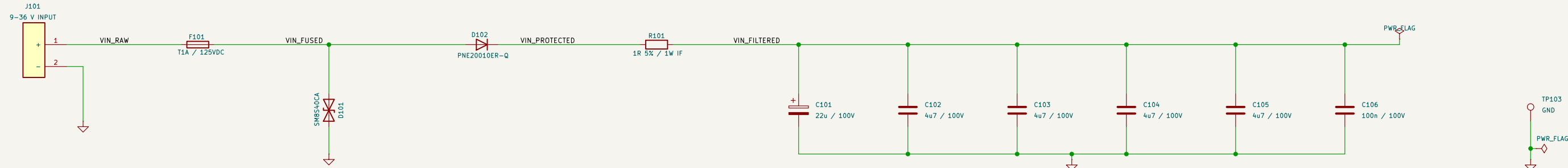
Title: Automotive BLE–CAN Module

Size: A4	Date: 2026–09–30	Rev: A
KiCad E.D.A. 10.0.6		Id: 1/5

01 AUTOMOTIVE POWER INPUT / VEHICLE-SUPPLIED RAILS

SCHEMATIC | 9–36 V nominal operating input

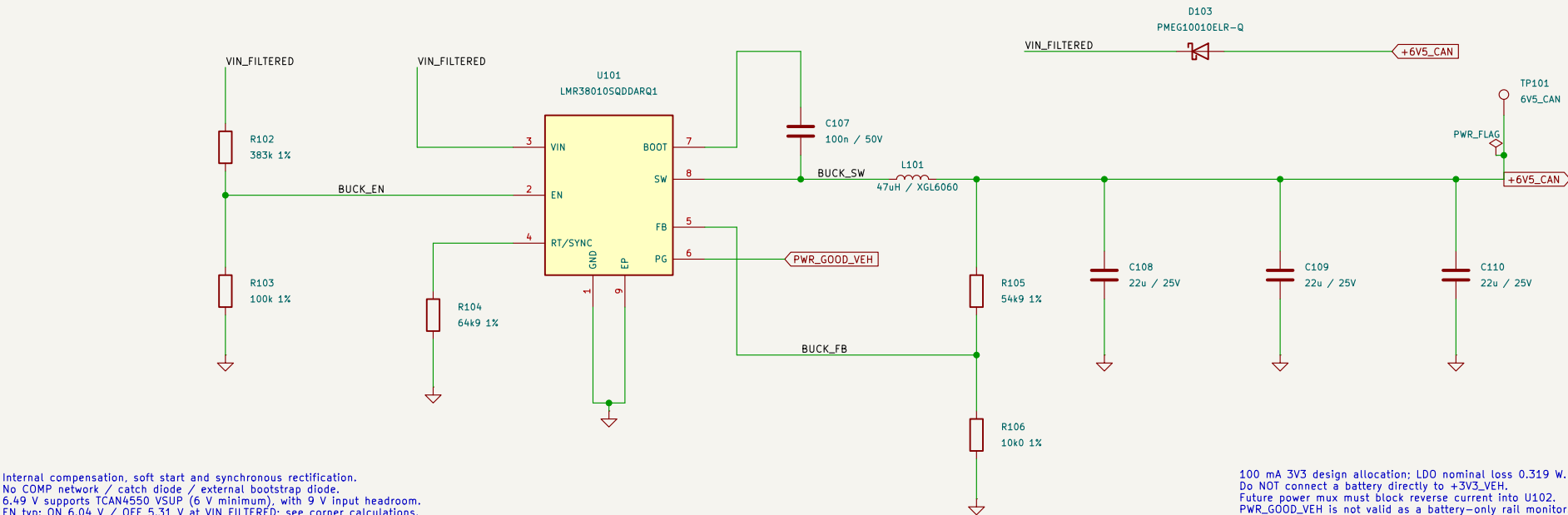
A. FUSED CLAMP + REVERSE BATTERY + DAMPED INPUT FILTER



F101 is NOT a precision current limiter; Use a suitably fused vehicle harness.
D101 return: short/wide to J101 return; keep surge current away from logic.
Bidirectional D101 avoids forward conduction on a reversed battery.

D101: 40 V stand-off; 64.5 V max clamp at 102 A, 10/1000 us, 25 degC.
80 V buck / 100 V input capacitors. Hot clamp + PCB overshoot must be checked.

B. 6.49 V SYNCHRONOUS BUCK / 300 mA TOTAL DESIGN BUDGET

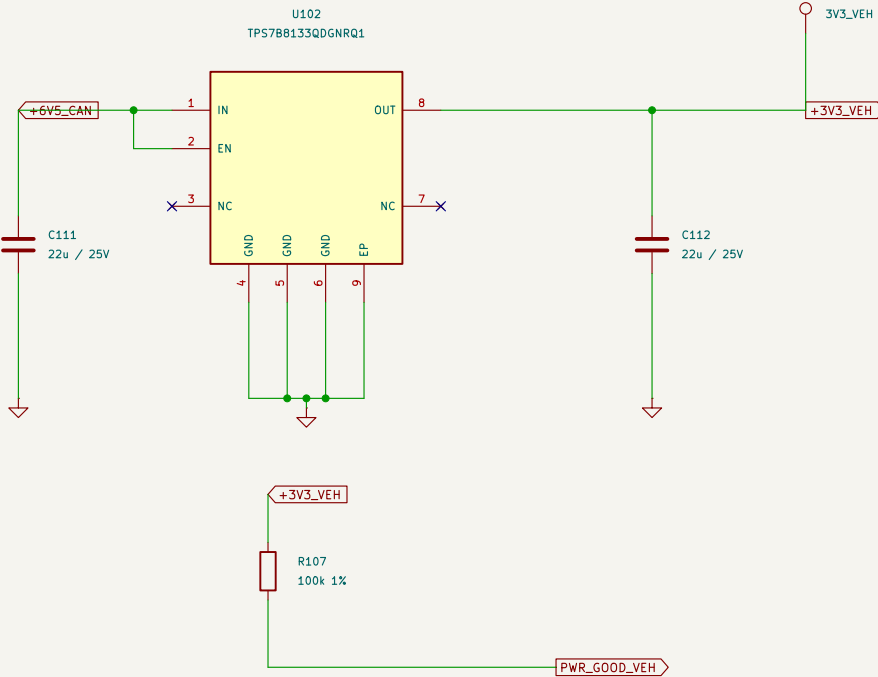


Internal compensation, soft start and synchronous rectification.
No COMP network / catch diode / external bootstrap diode.
6.49 V supports TCAN4550 VSUP (6 V minimum), with 9 V input headroom.
EN typ: ON 6.04 V / OFF 5.31 V at VIN_FILTERED; see corner calculations.
UVLO is not a guarantee of valid CAN operation during brownout.

100 mA 3V3 design allocation; LDO nominal loss 0.319 W.
Do NOT connect a battery directly to +3V3_VEH.
Future power mux must block reverse current into U102.
PWR_GOOD_VEH is not valid as a battery-only rail monitor.

Scope: only this power sheet. No +5V rail is required by the selected architecture.
Design is non-isolated. Never use GND as a substitute for a specified chassis connection.
No LED load added; test pads provide bring-up access without continuous drain.

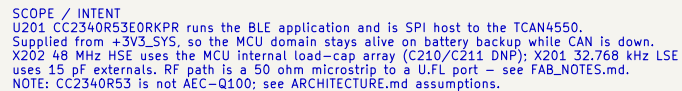
C. 3.3 V VEHICLE LOGIC SUPPLY



LAYOUT INTENT (PCB NOT IMPLEMENTED)
Keep C106 + ceramic bank at U101 VIN/GND; smallest hot loop.
Minimize SW copper; L101 marked short lead to SW.
Sense FB after L101 at output capacitors; away from SW/BOOT.
Connect both IC exposed pads to GND copper + thermal vias.
Keep a continuous ground plane; separate surge/switching returns by placement/routing, not by breaking the reference plane.
Target >=1.0 mm external-node clearance; verify fab / pollution / surge.

Design period: 2026–09–24 to 2026–09–30 Rev A		
4–layer FR4, 1.58 mm, 49.5 x 56.9 mm single-sided assembly		
CC2340R53 BLE TCAN4550–Q1 CAN FD 9–36 V input 1S Li-ion backup		
Automotive BLE–CAN Module		
SHIVAM TIWARI		
Sheet: /01_POWER_INPUT/		
File: 01_POWER_INPUT.kicad_sch		
Title: Automotive BLE–CAN Module		
Size: A3	Date: 2026–09–30	Rev: A
KiCad E.D.A. 10.0.6		Id: 2/5

SCHEMATIC | SPI host to TCAN4550 | U.FL antenna port



MCU PART NUMBER
CC2340R53E0RKPR (512 KB, QFN-40 RKP) is the selected
orderable device in the CC2340R5 family. This MCU is not
AEC-Q100 qualified. See the architecture and BOM notes.

REVIEW NOTES

C209 = 1uf on VDD0 (TI SWRA834A, section 9.1).

C210: AB506-3276KHZH-Z-T; CL=9pF; ESR<=90nOhm; -40 ~ +85C.

C212/C213: 156F GGG, 50V, 5%; GRM551HC50V0A01D;
EPCOS, C-Case, 156F; max temp. = 125C; tolerance on assembled PCB.

C210: ABM114-48.0000MHZ-7-DIX-T3; CL=7pF; ESR<=40nOhm, -40 ~ +85C.

C210/C211 DNP: use internal HFXT cap array; tune in SysConfig for this PCB.

D210: 0603 footprints provided; quality 100uA part current/DCR against TI reference.

C205/C208: select low QW parts with adequate effective capacitance at bias.

F201/F202: new design with FLZU01 capacitor with a continuous ground plane;

verify the added filter and UFL path by conducted RF testing.

Battery telemetry (VBAT_SENSE / USING_BATTERY_N) is not connected to the MCU yet.

SHIVAM TIWARI

Title: Automotive BLE-CAN Module

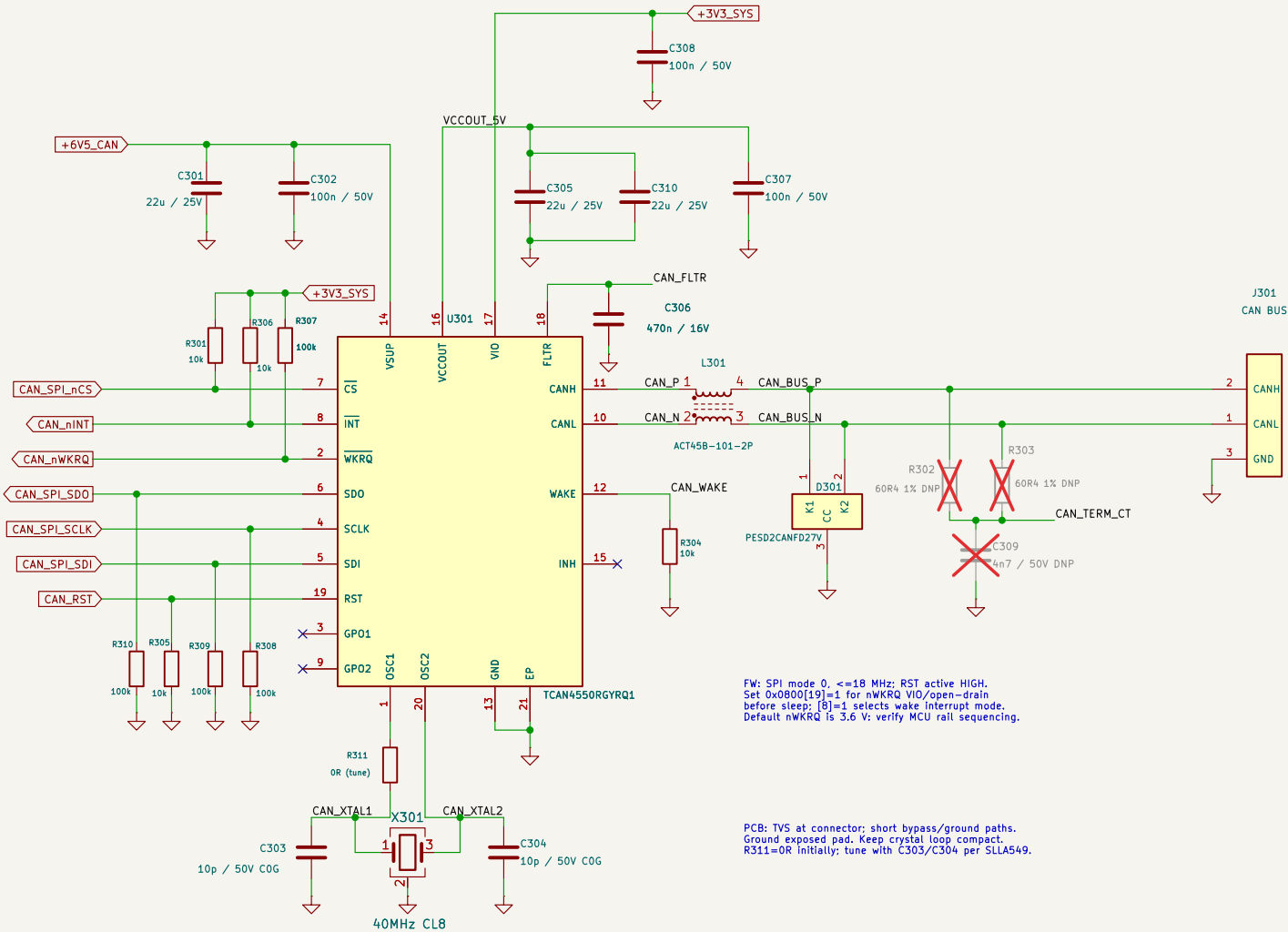
Rev: A

Id: 3/5

03 CAN INTERFACE / TCAN4550-Q1 CAN FD CONTROLLER + TRANSCEIVER

CAN INTERFACE - 3xx references / vehicle-powered CAN

R302/R303/C309: DNP for a stub. Populate together only at a bus end (120.8 ohm).



FW: SPI mode 0, <=18 MHz; RST active HIGH.
Set 0x0800[19]=1 for nWKRQ VIO/open-drain
before sleep; [8]=1 selects wake interrupt mode.
Default nWKRQ is 3.6 V; verify MCU rail sequencing.

PCB: TVS at connector; short bypass/ground paths.
Ground exposed pad. Keep crystal loop compact.
R311=0R Initially; tune with C303/C304 per SLA549.

C305+C310: >=10uF effective at 5V. C306: >=300nF effective. Verify MLCC bias/aging.
CAN operation requires valid VSUP and VIO >=3V; no CAN communication on backup-only power.

Design period: 2026-09-24 to 2026-09-30 | Rev A
4-layer FR4, 1.58 mm, 49.5 x 56.9 mm | single-sided assembly
CC2340R53 BLE | TCAN4550-Q1 CAN FD | 9-36 V input | 1S Li-ion backup
Automotive BLE-CAN Module

SHIVAM TIWARI

Sheet: /03_CAN_Interface/
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Title: Automotive BLE-CAN Module

Size: A4 | Date: 2026-09-30

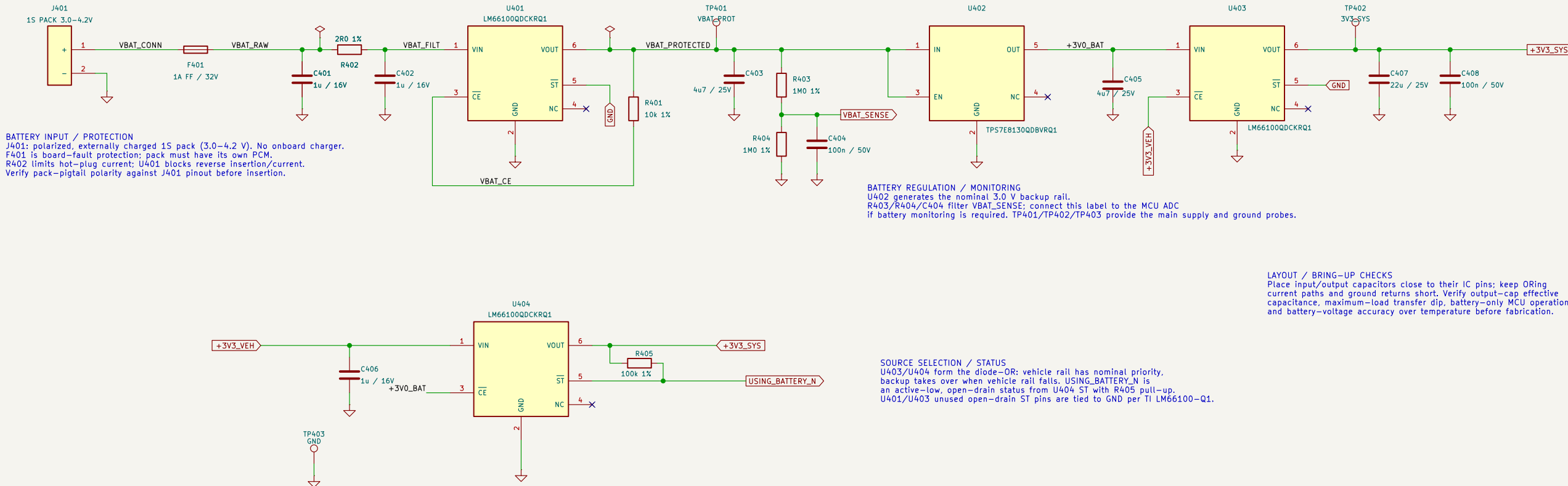
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Rev: A

Id: 4/5

04 BATTERY BACKUP / VEHICLE-BATTERY POWER PATH

SCHEMATIC REV A | replaceable externally-charged 1S pack | NO onboard charging | vehicle-priority diode-OR



Design period: 2026-09-24 to 2026-09-30 Rev A		
4-layer FR4, 1.58 mm, 49.5 x 56.9 mm single-sided assembly		
CC2340R53 BLE TCAN4550-Q1 CAN FD 9-36 V input 1S Li-ion backup		
Automotive BLE-CAN Module		
SHIVAM TIWARI		
Sheet: /04_Power_Back-up/ File: 04_Power_Back-up.kicad_sch		
Title: Automotive BLE-CAN Module		
Size: A3	Date: 2026-09-30	Rev: A
KiCad E.D.A. 10.0.6		Id: 5/5